

A Method for Identifying a Person Entering a Bathtub using a Water Pressure Sensor

Naoki Kurata
naoki.kurata@iis.ise.ritsumei.ac.jp
Ritsumeikan University
Shiga, Japan

Kazuya Murao
murao@cs.ritsumei.ac.jp
Ritsumeikan University
Shiga, Japan

ABSTRACT

It has been reported that the number of fatal accidents in bathtubs is particularly high in residential environments, so it is important to obtain information on body posture and other factors associated with individuals while bathing in bathtubs. In this study, we proposed a method to identify a person entering water by obtaining the change of water pressure from a water pressure sensor installed at the bottom of the bathtub in two situations; one is a situation in which a person is entering bathtub while pouring water into the bathtub, the other is a situation in which the bathtub overflows due to entering water. From evaluation experiments using entering water data collected from four subjects in pouring, overflow, and normal situations, an average F-value of 0.552 was obtained for the identification results in the overflow situation, and an average F-value of 0.792 was obtained for the identification results in the pouring situation.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing systems and tools.

KEYWORDS

Person identification, Water pressure sensor, Bathtub

ACM Reference Format:

Naoki Kurata and Kazuya Murao. 2022. A Method for Identifying a Person Entering a Bathtub using a Water Pressure Sensor. In *Proceedings of the 2022 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp/ISWC '22 Adjunct)*, September 11–15, 2022, Cambridge, United Kingdom. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3544793.3560389>

1 INTRODUCTION

With the miniaturization of computers, sensing at various locations has become possible, and the status and condition of the user can be easily recognized. Systems for sensing bio-information in residential environments such as bedrooms, bathrooms, and living rooms have been researched and developed so far. In addition, it is important to manage one's health condition on a regular basis

for the prevention and early detection of diseases. Among house environments, the bathtub is reported to have the highest number of fatal accidents [1]. 6,377 seniors died of unintentional drowning in 2014, of which 4,857 (76%) died in the bathtub [6]. In addition, approximately 4,800 seniors were reported to have died from bathtub drowning each year between 2011 and 2014 [6]. Therefore, it is particularly important to collect bio-information while bathing in a bathtub. The main cause of deaths in bathtubs is asphyxiation due to drowning caused by impaired consciousness, and it has been suggested that heat stroke, neurally mediated syncope caused by sitting in the bathtub, and arrhythmia are factors that can cause impaired consciousness [1]. The unique Japanese culture of bathing in hot water is also a background factor. In order to prevent such fatalities, it is important to understand the condition of person entering water.

Information that can be obtained in the bathroom includes the bather's personal identification information, bio-information such as the bather's heartbeat, and bathing information such as the time for taking a bath and the amount of used hot water. The bio-information and bathing information need to be linked to the individual and managed, and we believe it is useful to simply record who take a bath. Therefore, the goal of this research is to identify the person who enter the bathtub.

Based on these considerations, the research group to which the authors belong has previously investigated a method to identify bathers based on changes in the hot water level during bathing by installing a water pressure sensor at the bottom of the bathtub [12]. The contribution of [12] is that it can identify individuals with high accuracy while not restricting their bathing posture. However, the system assumes only a constant and unchanging amount of hot water in the bathtub while bathing, and doesn't assume a situations in which users enter the bathtub while pouring hot water or hot water overflow when the they enter the bathtub, which is not realistic. Therefore, in this study, we propose a method to identify the person entering water while pouring hot water into the bathtub and when hot water overflow from the bathtub due to users entry the bathtub.

2 RELATED WORK

In this section, we introduce research on behavior recognition and personal identification in residential environments and research on behavior recognition and personal identification in bathrooms.

Guo et al. [3] proposed a system that measures the respiration and heartbeat of users lying on a bed with a tactile sensor. Using the tactile sensor, the system extracts respiration and heartbeat from the collected time-series data of pressure values through signal processing based on differences in frequency bands. Tanaka et al. [17]

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

UbiComp/ISWC '22 Adjunct, September 11–15, 2022, Cambridge, United Kingdom

© 2022 Association for Computing Machinery.

ACM ISBN 978-1-4503-9423-9/22/09...\$15.00

<https://doi.org/10.1145/3544793.3560389>

proposed a system to measure the cycle of respiration and heartbeat during sleep by attaching a PCG sensor to the side of an air mat. In places such as restrooms and bathrooms, it is important to protect the user's privacy as well as the conditions that isn't burden to the user, such as non-contact and non-restraint. Namatame et al. [13] proposes Life2Guard, a system for detecting physical abnormalities in restrooms. Using multiple infrared sensors installed on the wall and an acceleration sensor installed on the toilet seat, the system detects abnormal behavior or motionlessness in the toilet. In addition, Kurahashi et al. [10] proposed a method to identify restroom users based on the rotation speed and usage of toilet paper by attaching an angular velocity sensor to the toilet paper holder. Fogarty et al. [8] proposed a system to recognize the use of a washing machine or restroom based on the sound of running water in a drainpipe by installing a low-cost device using a microphone outside the drainpipe. Since the device is installed inside the drain pipe, it is not visible to the public and privacy is protected, but it does not personally identify the water user. In addition, several methods have been proposed to attach pressure sensors under the four legs of a desk to recognize desk movements and identify desk users based on changes in the position of the gravity center on the top board [9][16][2].

Yoshimura et al. [21] proposed a system to measure electrocardiogram (ECG) by placing electrodes in a bathtub as a means of sensing bio-information in a bathtub. Kurihara et al. [18] proposed a system for sensing bio-information on the floor or in a bathtub by connecting an ultra-sensitive pressure sensor to a silicon tube sandwiched between two plates and measuring changes in air pressure in the tube. Obo et al. [14] propose a system to sense the user's bio-information by placing an ultra-sensitive vibration sensor on the edge of the bathtub. The system detects the micro-vibrations of the user's heartbeat and respiration propagating through the bathtub while the user's back is in contact with the bathtub. These methods require the user to touch the bathtub while bathing, which is an operation unrelated to bathing and may be a burden to the user.

In homes and facilities, the bathtub is used by multiple people. Therefore, it is necessary to classify the bio-information obtained during bathing by each user. Winarno et al. [19] have proposed a facial recognition-based attendance system by using a camera as personal identification using features of each user. However, installing a camera in the bathroom is inappropriate due to privacy issues and the possibility of causing discomfort to the user.

As a method that ensures privacy and does not require the user to perform operations unrelated to bathing, Xu et al. [20] have proposed a system that identifies the user from the ECG obtained. The system can measure and identify the user without attaching sensors to the user's body, but the user cannot change user's body position freely because the ECG measured changes when the position of the body and electrodes change. Ince et al. [7] recognizes actions such as washing face, brushing teeth, etc. using pressure sensors attached to the environment and wrist-mounted accelerometers. Hirai et al. propose [5], a system that plays music based on the values obtained from the heart rate measuring electrodes installed on the wall of the bathtub and the water pressure sensor installed in the water heater, and [4], a system that can be used as an interface

by embedding a electrostatic capacity touch sensor in the edge of the bathtub.

In this study, we propose an unconscious and unconstrained method for identifying person entering water using water level change information that does not impose a psychological burden on the person entering the water and does not require person entering water to take a specific posture.

3 PROPOSED METHOD

This section describes a method to identify a person entering water in the situation where hot water is being poured and hot water overflows due to bathing based on the time-series change of the water pressure (water level) in the bathtub using a water pressure sensor installed at the bottom of the bathtub.

3.1 Assumed Environment

We believe that the water pressure sensor does not infringe on the user's privacy because it acquires only the water pressure on the sensor part, i.e., the depth of the bathtub. In addition, the water pressure sensor can acquire water pressure data simply by entering the water as usual, without imposing any restrictions on how to enter the water, such as a specific posture or contact with the electrodes.

The time series of the water pressure sensor when a person enters the bathtub is shown in Figure 1. The horizontal axis is the time, and the vertical axis is the AD converted output voltage value of the water pressure sensor. In this study, as shown in Figure 1, we define "entering water" as the interval from the moment when the user touches the hot water in the bathtub and the water level rises to the moment when the body is immersed in the hot water in the bathtub and the water level stabilizes, and "exiting water" as the interval from the moment when the user gets out of the hot water in the bathtub and the water level decreases to the moment when all the body parts are out of the water and the water level stabilizes. The interval from the end of entering water to the start of exiting water is defined as "bathing". A series of entering water, bathing, and exiting water intervals is called a single "bath". A situation in which a person is entering bathtub while pouring water into the bathtub is called a "pouring situation," a situation in which the bathtub overflows due to entering water is called a "overflow situation," and a situation in which is not pouring the water into the bathtub and does not overflow due to entering water is called a "normal situation".

As a constraint, the number of people in the bathtub at one time is assumed to be one in this study. The measurement error of the initial amount of hot water in the bathtub is assumed to be within 2 cm in each of the three situations: normal, pouring, and overflow. In other words, the amount of hot water before entering water is almost the same in the normal situation and the measurement error is within 2cm, but the amount of hot water before entering water in the three situations is not the same.

3.2 Hardware Configuration

The proposed system uses a waterproof box that contains a water pressure sensor (HTV-020KP-02-V by Sensez company), Arduino Uno, SD card and SD card slot, 15V power supply (10 AA battery),

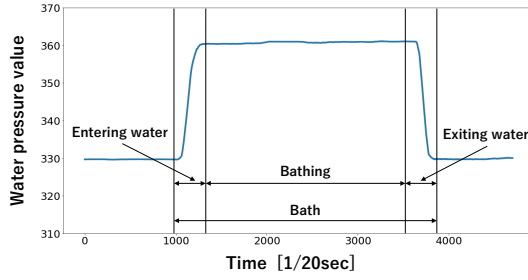


Figure 1: Water pressure sensor values when entering exiting water.

and waterproof on/off switch. Install the box outside the bathtub, and extend the measuring part of the water pressure sensor outside the box and fix it to the bottom of the bathtub with waterproof tape. The waterproof on/off switch is exposed outside the box, allowing the device to be turned on and off without opening and closing it. When the power is turned on, the device automatically starts sensing and records the water pressure value to the SD card. The sampling frequency of the water pressure value is 50 Hz. This water pressure sensor acquires gauge pressure, with a pressure rating of 0–20 kPa and an output signal range of 1–5 V. The gauge pressure is the absolute pressure minus the atmospheric pressure. Since the water pressure at a depth of 1 cm is 100 Pa¹, This water pressure sensor can measure a range of water depths up to 0–200 cm. The used Arduino Uno microcontroller board has an operating voltage of 5 V and analog input pins with 10 bit resolution, so the water depth can be measured with a resolution of about 2 mm. At the present stage, the installed device acquires sensor data of bath, records it in the device’s SD card, and identifies the user by analyzing the data offline. We install only a water pressure sensor extended from the box inside the bathtub, so it doesn’t interfere with the usual bathing position.

3.3 Outline of the Method

The flow of the method for identifying people entering water is shown in Figure 2. In this study, the sensor data stored on the SD card in the device is analyzed offline by a PC. This is a limitation of the hardware, and the proposed method is an algorithm that can be processed online by transmitting the acquired data in real time. First, the situation in the bathtub at the time of water entry is determined. Then, the waveform of one water entry section is segmented from the acquired water pressure data. If the result of the identification of the situation in the bathtub is the pouring situation, the changes in water pressure due to pouring water are removed and the changes in water pressure due to entering water only are obtained. After that, the DTW distance between the entering water waveform, in which the change in water pressure due to pouring water is removed, and the entering water waveform under the normal situation, which has been registered in advance with the individual, is calculated. Finally, the calculated DTW distance is used to determine the most similar

¹ 1 Pa = 1 N/m² and the density of water is 1 g/cm³, the force on 1 m² at a depth of 1 cm is 100Pa.

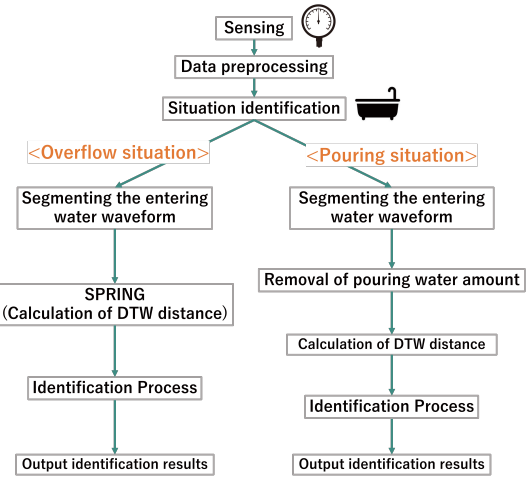


Figure 2: Outline of the proposed method.

person using the k-nearest neighbor method, and the resulting individual in the registration data is output as the identification result. If the result of the identification of the situation in the bathtub is the overflow situation, by using SPRING, we calculate the DTW distance to the entering water waveform under the normal situation, which has been registered in advance with the individual. Finally, the calculated DTW distance is used to determine the most similar person using the k-nearest neighbor method, and the resulting individual in the registration data is output as the identification result.

3.4 Data Preprocessing

If the water pressure value acquired at time t is $p(t)$, $p(t)$ contains the noise of the water surface oscillation generated during bathing. The water pressure value $p_f(t)$ is obtained by removing the noise using the average value filter shown in the following equation. However, n is the window width of the filter, which is set to $n = 200$ in this study.

$$p_f(t) = \frac{1}{n} \sum_{i=0}^{n-1} p(t-i) \quad (1)$$

Since the goal of this study is to identify individuals when they enter the water, it is necessary to segment the waveform of the section where they are entering water from the acquired water pressure data $p_f(t)$. For this purpose, we detect the start time of entering water, which is the moment when the body touches the water in the bathtub and the water level rises, and the end time of entering water, which is the moment when the body is immersed in the hot water in the bathtub and the water level stabilizes. After that, the water pressure data from the start time of the entering water to the end time of the entering water is defined as the entering water waveform.

The first step is to detect the start time of entering water. In this study, the water pressure increases monotonically in the pouring situation. However, when a user enters the water, the water pressure value increases more than the amount of increase caused by pouring

water. In order to detect the sudden increase in water pressure when a user enters the water, the deviation value $p_d(t)$ between the current value $p_f(t)$ and the moving average value $p_a(t)$ is calculated according to the following equation. However, m is the window size of the moving average, and in this study, $m = 200$.

$$p_a(t) = \frac{1}{m} \sum_{i=0}^{m-1} p_f(t-i) \quad (2)$$

$$p_d(t) = p_f(t) - p_a(t) \quad (3)$$

3.5 Situation Identification

In order to detect the start time of entering water, it is necessary to distinguish between three types of situations: normal, pouring, and overflow. First, we observe the deviation values of the last 100 samples $p_d(t-i)$ ($i = 0, \dots, 99$) at time t . If all the deviation values $p_d(t-i)$ of the 100 observed samples satisfy $h_L < p_d(t-i) < h_H$ ($h_L = 0.2$, $h_H = 3.0$), we judge that it's in the pouring situation, and determine the start and end time of entering water in the pouring situation. If this condition is not satisfied, we judge that it's not in the pouring situation and observe the water pressure values $p_f(t-i)$ ($i = 0, \dots, 9$) of the last 10 samples at time t . If all the water pressure values $p_f(t-i)$ of the 10 samples we observed satisfy $p_f(t-i) > h_M$ ($h_M = 427.5$), we judge that it's in the overflow situation and determine the start and end time of entering water in the overflow situation. If neither of the conditions is satisfied, we judge it's in the normal situation. If the situation is judged to be the normal, the entering water waveform in the normal situation is segmented using the method shown in Section 3.7.1 and used as training data.

3.6 In the pouring situation

The following is the process to be followed when the pouring situation is determined.

3.6.1 Segmenting the entering water waveform in the pouring situation. The following is the process for segmenting the entering water waveform in the pouring situation. The deviation value $p_d(t)$ when a user enters the water in the pouring situation is shown in Figure 3. The deviation value $p_d(t)$ is used to detect the sudden increase in water pressure value that occurs when a user enters the water and to determine the start time of entering water. In this study, we first detect the time t_u when $p_d(t) > h_1$ ($h_1 = 7.0$) is satisfied for the first time in 50 consecutive samples, as shown in Figure 3. The time t when the deviation value $p_d(t)$ in the immediate past falls below the threshold h_2 ($h_2 = 3.0$) for the first time is set as the start time of entering water t_s . In this study, if the deviation value $p_d(t)$ exceeds h_1 , but falls below h_1 within 50 samples, it is judged to be a change in water level due to posture shift during bathing, and the time t is searched again so that $p_d(t) > h_1$ for 50 consecutive samples. Then, the end time of entering water is detected, and the waveform of the entering water section is segmented. The end time of entering water t_e is defined as the time t when the deviation value $p_d(t)$ from the start time of entering water t_s in the immediate future falls below the threshold h_2 for the first time. Then, the water pressure data $p_f(t)$ from the start time of entering water t_s

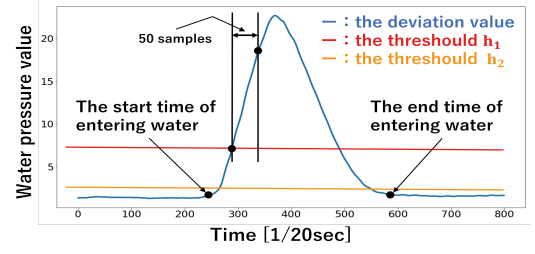


Figure 3: Deviation value of entering water section in the pouring situation.

to the end time of entering water t_e is acquired as data of entering water section $p_{in}(t) = [p_f(t_s), \dots, p_f(t_e)]$.

3.6.2 Removal of pouring water amount. In the pouring situation, the entering water waveform differs depending on the amount of pouring water. Therefore, when it is judged to be the pouring situation, the effect of pouring water amount is subtracted from the entering water waveform to obtain only the water pressure change caused by entering water. This process is not performed for overflow and normal conditions. The pouring water amount is estimated by calculating the regression line $p_p(t)$ based on the least squares method using the water pressure values of the last 100 samples at the start time t_s . In the case of no pouring water, the pouring water amount is estimated to be almost zero, so the method can be applied to the overflow situation as well. When the regression line to be estimated is $y = ax + b$, it is calculated using the following equation.

$$a = \frac{\sum_{n=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{n=1}^n (x_i - \bar{x})^2} \quad (4)$$

$$b = \bar{y} - a\bar{x} \quad (5)$$

By subtracting the pouring water amount $p_p(t)$ calculated using the above method from the entering water waveform $p_{in}(t)$, we obtain the entering water waveform $p_e(t)$ without the effect of the pouring water amount.

$$p_e(t) = p_{in}(t) - p_p(t) \quad (6)$$

This entering water data $p_e(t)$ is used as input data for identification.

3.6.3 Calculation of DTW distance. We calculate the similarity between the entering water data without the effect of pouring water amount $p_e(t)$ and the training data $p_r(t)$ collected in advance under the normal conditions. After that, the personal label assigned to the training data with the highest similarity is output as the identification result. A commonly used method for calculating similarity is the Euclidean distance. However, Euclidean distance is susceptible to time distortions in time series data and requires the same number of samples for comparison. DTW (Dynamic Time Warping) is a similarity calculation method for time series data that overcomes the disadvantages of the Euclidean distance [11]. A feature of DTW is that it can calculate the similarity between data with different time series scales and different number of samples by computing the temporal nonlinear compression distance.

3.6.4 Identification process in pouring situation. The name of the recognition target is linked to the entering water waveform collected from multiple recognition targets in advance under normal situation, and is stored as training data $p_r(t)$. In order to eliminate the effect of the amount of water in the bathtub and to consider only the increase or decrease of the water level due to entering water, all entering water waveform was translated so that its minimum value became zero. For identification in the pouring situation, the entering water waveform $p_e(t)$ obtained in Section 3.6.2 is used as input data. Compute the DTW distance between the input data $p_e(t)$ and all training data $p_r(t)$, and determine the identification result by majority vote of the labels assigned to the training data showing the k-nearest neighbor DTW distances.

3.7 In the overflow situation

The following is the process to be followed when the overflow situation is determined.

3.7.1 Segmenting the entering water waveform in the overflow situation. The following is the process for segmenting the entering water waveform in the overflow situation. In the overflow situation, the deviation value $p_d(t)$ is close to zero because the water pressure value does not increase monotonically at times other than when the user enters or leaves the water, unlike in the pouring situation. In this study, we first detect the time t_u when $p_d(t) > h_1$ is satisfied for the first time in 50 consecutive samples. The time t when the deviation value $p_d(t)$ in the immediate past falls below the threshold h_3 ($h_3 = 0.20$) for the first time is set as the time when water entry starts t_s . In the same way as for pouring situation, if the deviation value $p_d(t)$ exceeds h_1 , but falls below h_1 within 50 samples, it is judged to be a change in water level due to posture shift during bathing, and the time t is searched again so that $p_d(t) > h_1$ for 50 consecutive samples. Then, the end time of entering water is detected, and the waveform of the entering water section is segmented. The end time of entering water t_e is defined as the time t when the deviation value $p_d(t)$ from the start time of entering water t_s in the immediate future falls below the threshold h_3 for the first time. Then, the water pressure data $p_f(t)$ from the start time of entering water t_s to the end time of entering water t_e is acquired as data of entering water section $p_{in}(t) = [p_f(t_s), \dots, p_f(t_e)]$. If it is judged to be normal situation, the process is the same as that for overflow situation. The proposal of a process that distinguishes between overflow and normal situations is a future subject. If the situation is judged to be the normal, the same segmenting process as for the overflow situation is also used.

3.7.2 SPRING (calculation of DTW Distance). We calculate the similarity between the entering water data in the overflow situation $p_{in}(t)$ and the training data $p_r(t)$ collected in advance under the normal conditions. The initial water level is different between the overflow situation and the normal situation, so the waveform is different for the same subject entering water. Therefore, we use SPRING [15], which is a DTW-based method for extracting subsequences similar to the query sequence from the data stream. By using SPRING, it is possible to identify person entering water by extracting partially similar sequence even when they enter the water under different initial water level.

3.7.3 Identification process in overflow situation. The name of the recognition target is linked to the entering water waveform collected from multiple recognition targets in advance under normal situation, and is stored as training data $p_r(t)$. In order to eliminate the effect of the amount of water in the bathtub and to consider only the increase or decrease of the water level due to entering water, all entering water waveform was translated so that its minimum value became zero. For identification in the overflow situation, the entering water waveform $p_{in}(t)$ obtained in Section 3.7.1 is used as input data. We calculate the training data with the subsequences that satisfy $DTW(p_{in}(t), p_r(t)) < T_h$ to the test data from among all training data, and determine the identification result by majority vote of the labels assigned to the training data showing the k-nearest neighbor DTW distances. However, T_h is the threshold for detecting the subsequence, and in this study, $T_h = 1000$.

4 EVALUATION

This section describes the experiments conducted to evaluate the accuracy of the identification of person entering water in order to verify the effectiveness of the proposed method.

4.1 Evaluation Environment

We installed the device shown in Section 3.2 in the bathroom of a hotel room in Shiga prefecture, and obtained water level data from four male subjects in their 20s (subjects A, B, C, and D) when they actually entered the bathtub. In this study, we conducted evaluation experiments in two situations: pouring and overflow situation. Each subject repeatedly bathed 10 times as a set of entering and leaving in each of two situations, and collected water pressure data for 80 times in total from four subjects. These data were used as input data. As training data, each subject repeatedly bathed 15 times as a set of entering and leaving water under the normal conditions, and collected water pressure data for 60 times in total from four subjects. In the data collection, we asked the subjects to wait in the bathroom for about 2 minutes after leaving water until the next time entering water in order to suppress the oscillation of the water surface caused by leaving water. We did not give any special instructions on how to bath, or on the direction of the body and movements during bathing, and the subjects were free to perform as usual. We calculated the DTW distance between the entering water waveform of the training data collected under the normal situation and the entering water waveform collected in the pouring situation or the overflow situation by using the method showed in Section 3.6.3, Section 3.7.2. The calculated DTW distance was used to output the identification result of person entering water by majority vote of the labels of the training data based on the k-nearest neighbor method ($k = 5$).

4.2 Results and Discussion in the Overflow Situation

The confusion matrix of the identification results obtained based on the k-nearest neighbor method ($k = 5$) using the calculated DTW distance is shown in Table 1, and the recall, precision, and F-value are shown in Table 2. In the confusion matrix, the rows show the correct labels and the columns show the identification results. Each cell shows the number of samples of the identification results for

Table 1: Confusion matrix of identification results for each subject in the overflow situation.

Input \ Output	Subject A	Subject B	Subject C	Subject D
Subject A	10	0	0	0
Subject B	3	7	0	0
Subject C	0	10	0	0
Subject D	2	0	0	8

Table 2: Recall, Precision, and F-value for each subject in the overflow situation

	Recall	Precision	F-value
Subject A	1.000	0.667	0.800
Subject B	0.700	0.412	0.519
Subject C	0.000	0.000	0.000
Subject D	0.800	1.000	0.889
Average	0.625	0.520	0.552

the correct labels, and the total number of samples in the rows is 10. The diagonal component from the upper left to the lower right shows the result of correct classification.

From the results of Table 2, the F-values of subjects A, B and D were 0.800, 0.519 and 0.889 but the F-values of subject C was 0.000, resulting in an average of 0.552. In the overflow situation, the initial water level was set higher than the normal situation in order to make the water overflow by entering water for all subjects.

When entering the water in the overflow situation, if the amount of overflowing water is small, even the shoulder and neck, which are not immersed in the water in the normal situation, will be immersed in the water. If the amount of overflowing water is large, the water overflow when only the waist is immersed, and the water pressure data becomes constant thereafter, and the data of the body parts cannot be obtained thereafter. The entering water data in the overflow situation acquired this time included the data of the body parts that were not included among the entering water data in the normal situation like the former. For this reason, it is considered that subject C's data became similar to subject B's objects. Therefore, we believe that it was difficult to identify them with high accuracy.

4.3 Results and Discussion in the Pouring Situation

The confusion matrix of the identification results obtained based on the k-nearest neighbor method ($k = 5$) using the calculated DTW distance is shown in Table 3, and the recall, precision, and F-value are shown in Table 4.

From the results of Table 4, the F values of subjects A and D were 0.946 and 1.000, respectively, while the F values of subjects B and C were 0.571 and 0.600, respectively, in an average of 0.792. Subjects B and C had similar body sizes compared to the other subjects, and there was a lot of variability in their entering water behavior each time their bathing, which may have reduced the identification accuracy. By conducting additional experiments and collecting more entering water data in the normal situation, we believe we can

Table 3: Confusion matrix of identification results for each subject in the pouring situation.

Input \ Output	Subject A	Subject B	Subject C	Subject D
Subject A	9	1	0	0
Subject B	0	6	4	0
Subject C	0	4	6	0
Subject D	0	0	0	10

Table 4: Recall, Precision, and F-value for each subject in the pouring situation.

	Recall	Precision	F-value
Subject A	0.900	1.000	0.946
Subject B	0.600	0.545	0.571
Subject C	0.600	0.600	0.600
Subject D	1.000	1.000	1.000
Average	0.775	0.786	0.792

identify a variety of entering water movements. On the other hand, the subjects of this experiment were four males in their twenties, who are considered to have similar physical characteristics and agility of movement. In addition, considering the use of the system at home, the data that can be collected will be diverse due to the differences in gender and age, and higher identification accuracy may be obtained.

5 CONCLUSION

In this study, we proposed a method to identify a person entering water by obtaining the change of water pressure from a water pressure sensor installed at the bottom of the bathtub in two situations; one is a situation in which a person is entering bathtub while pouring water into the bathtub, the other is a situation in which the bathtub overflows due to entering water. In the overflow situation, an average F-value of 0.552 was obtained as a result of an evaluation experiment using entering water data collected from four subjects a total of 40 times. In the overflow situation, if the amount of overflowing water is small, the shoulders and neck, which are not immersed in water in the normal situation, will be immersed in water. In this study, the entering water data in the overflow situation included data of body parts that were not included among the entering water data in normal situation. For this reason, we believe that it was difficult to identify them with high accuracy. In the pouring situation, an average F-value of 0.792 was obtained from the evaluation experiment using the entering water data collected from four subjects a total of 40 times. Considering the experimental environment in which the subjects' age, gender, and body size are close, the results suggest that the proposed method has a certain effectiveness as a method to identify person entering water in the pouring situation.

REFERENCES

- [1] Haruhiko Abe, Ritsuko Kohno, and Yasushi Oginosawa. 2013. Reflex syncope during a hot bath as a specific cause of drowning in Japan. *Journal of Arrhythmia* 29, 1 (02 2013), 37–38.

- [2] A.Schmidt, M.Strohhach, K.van Laerhoven, and H.-W Gellersen. 2003. Ubiquitous Interaction - Using Surfaces in Everyday Environments as Pointing Devices. *7th ERCIM Workshop* (2003), 263–279.
- [3] Shijie Guo, Xingli Zhao, Kazuya Matsuo, Jinyue Liu, and Toshiharu Mukai. 2019. Unconstrained Detection of the Respiratory Motions of Chest and Abdomen in Different Lying Positions Using a Flexible Tactile Sensor Array. *IEEE Sensors Journal* 19, 21 (2019), 10067–10076. <https://doi.org/10.1109/JSEN.2019.2925022>
- [4] Shigeyuki Hirai and Daiki Ito. 2015. Entertainment Applications for Tapping on a Bathtub Edge Using Embedded Acoustic Sensors. (11 2015). <https://doi.org/10.1145/2832932.2832983>
- [5] Shigeyuki Hirai, Yoshinobu Sakakibara, and Seiho Hayakawa. 2012. Bathcratch: Touch and Sound-Based DJ Controller Implemented on a Bathtub. (2012), 44–56.
- [6] Wan-Hua Hsieh, Chien-Hsing Wang, and Tsung-Hsueh Lu. 2019. Bathtub drowning mortality among older adults in Japan. *International Journal of Injury Control and Safety Promotion* 26, 2 (2019), 151–155. <https://doi.org/10.1080/17457300.2018.1515231> PMID: 30239269.
- [7] Nuri Firat Ince, Cheol-Hong Min, Ahmed Tewfik, and David Vanderpool. 2008. Detection of Early Morning Daily Activities with Static Home and Wearable Wireless Sensors. *EURASIP J. Adv. Signal Process* 2008, Article 31 (jan 2008), 11 pages. <https://doi.org/10.1155/2008/273130>
- [8] J.Fogarty, C.Au, and S.E.Hudson. 2006. A Feasibility Study of Unobtrusive and LowCost Home Activity Recognition. *UIST 2006* (2006), 91–100.
- [9] K.Murao, J.Imai, T.Terada, and M.Tsukamoto. 2017. Activity Recognition and User Identification based on Tabletop Activities with Load Cells. *Journal of Information Processing* 25, 1 (2017), 59–66.
- [10] Masaya Kurahashi, Kazuya Murao, Tsutomu Terada, and Masahiko Tsukamoto. 2017. Personal identification system based on rotation of toilet paper rolls. (03 2017), 521–526. <https://doi.org/10.1109/PERCOMW.2017.7917617>
- [11] Guiling Li, Yuanzhen Wang, Min Li, and Zongda Wu. 2008. Similarity Match in Time Series Streams under Dynamic Time Warping Distance. 4 (2008), 399–402. <https://doi.org/10.1109/CSSE.2008.1117>
- [12] Kazuya Murao, Sarina Nakayama, Masahiro Mochizuki, and Nobuhiko Nishio. 2018. User Identification Method in a Bathtub with a Water Pressure Sensor. (2018), 133–139. <https://doi.org/10.1145/3282353.3282360>
- [13] Naoya Namatame, Jin Nakazawa, Kaznori Takashio, and Hideyuki Tokuda. 2008. Life2Guard: A Physical Disorder Detection in Private Rooms. (07 2008), 177 – 180. <https://doi.org/10.1109/SAINT.2008.86>
- [14] Takenori Obo, Toshiyuki Sawayama, Takuya Sawayama, and Naoyuki Kubota. 2017. Fuzzy echo state network for heartbeat detection using ultrasensitive vibration sensor. (2017), 1–6. <https://doi.org/10.1109/IFSA-SCIS.2017.8023220>
- [15] Yasushi Sakurai, Christos Faloutsos, and Masashi Yamamuro. 2007. Stream Monitoring under the Time Warping Distance. *Proceedings - International Conference on Data Engineering* (04 2007), 1046–1055. <https://doi.org/10.1109/ICDE.2007.368963>
- [16] Albrecht Schmidt, Martin Strohhach, Kristof Van Laerhoven, Adrian Friday, and Hans Gellersen. 2002. Context Acquisition Based on Load Sensing. *LNCS* (09 2002), 333–350. https://doi.org/10.1007/3-540-45809-3_26
- [17] Wakimoto K. Tanaka S, Matsumoto Y. 2002. Unconstrained and non-invasive measurement of heart-beat and respiration periods using a phonocardiographic sensor. *Med Biol Eng Comput.* 40, 2 (2002), 246–252. <https://doi.org/10.1109/JSEN.2019.2925022>
- [18] Kaji Watanabe, Yosuke Kurihara, and Hiroshi Tanaka. 2009. Ubiquitous Health Monitoring at Home – Sensing of Human Biosignals on Flooring, on Tatami Mat, in the Bathtub, and in the Lavatory. *IEEE Sensors Journal* 9, 12 (2009), 1847–1855. <https://doi.org/10.1109/JSEN.2009.2030987>
- [19] Edy Winarno, Imam Amin, Herny Februriyanti, Prajanto Adi, Wiwien Hadikurniawati, and Muchamad Anwar. 2019. Attendance System Based on Face Recognition System Using CNN-PCA Method and Real-time Camera. (12 2019), 301–304. <https://doi.org/10.1109/ISRITI48646.2019.9034596>
- [20] Jianbo Xu, Tianhui Li, Peng Cui, and Wenxi Chen. 2019. Improvement of ECG based Personal Identification Performance in Different Bathtub Water Temperature by CNN. (10 2019), 1–5. <https://doi.org/10.1109/ICAWS.2019.8923503>
- [21] Takumi Yoshimura, Kazuki Nakajima, Hidetoshi Miike, Toshiyo Tamura, and Tatsuo Togawa. 1994. Development and Evaluation of Heart Rate Monitor at Bathtub. *Japanese journal of medical electronics and biological engineering* 32, 4 (1994), 246–253. <https://doi.org/10.11239/jsmbe1963.32.246>